

Annexes

The 10-step MAP Procedure

-
- Step 1. Prepare a fully written Standard Operating Procedure (SOP) and define the analyte exactly, i.e. the quantity being measured.
 - Step 2. Select the bounds of the validation range and define the acceptance interval, which can be variable with the concentration.
 - Step 3. Select at least three *validation materials* representative of the validation range and the analyzed matrix. They are named *levels* and their exact contents must be known (they are the reference values) with an eventual uncertainty. They can be certified materials, surrogates, natural, or spiked matrices.
 - Step 4. Define the validation experimental design in terms of the number of *series* and replicates/per series. This is a crucial step that may condition the success of the study (see Section 5.4.2).
 - Step 5. If the method requires an external calibration, define the calibration experimental design.
 - Step 6. Run assays, collect measurements according to planned designs, and make a note of any discrepancies.
 - Step 7. If the method requires external calibration, estimate calibration curves for each series and calculate inverse-predicted concentrations.
 - Step 8. For each level (or validation material), calculate validation parameters, i.e. intermediate precision standard deviation and bias, with inverse-predicted concentrations.
 - Step 9. For each level, calculate the statistical dispersion interval containing the expected proportion of measurements as explained in Section 5.3.
 - Step 10. Interpret the results and define the validated range of the method, if any.
-

Glossary of Used Terms

The vocabulary relating to the performance or validation of analytical methods is very abundant often confusing. For the purposes of this book, many definitions are derived from the International Vocabulary of Metrology published by the BIPM [1], where they exist. Some of the relevant definitions come from other documents and are given succinctly. Supplementary information can be found in the normative documents cited at the end of each chapter.

Term	Definition
Acceptance interval	Interval of permissible measured values. Unless otherwise stated in the specification, the acceptance limits belong to the acceptance interval
Acceptance limit	Specified upper or lower bound of permissible measured values
Accuracy	Closeness of agreement between a measured value and a true value of a measurand
Accuracy profile	Combination, in graphical form, of one or more statistical dispersion intervals calculated at different concentration levels and one or more acceptance intervals
Analyte	Term used in analytical sciences instead of measurand
Between-series variance	Refers to the influence on the outcome of the organization of measures in the form of identifiable groups (related to ANOVA)
Bias	Estimate of a systematic error
Bound	Limit of an interval
Calibrant Surrogate Standard	Chemical compound used as a substitute for the authentic analyte for preparing calibration standards or calibrators
Calibration	Operation that, under specified conditions, in the first step, establishes a relation between the values with measurement uncertainties provided by measurement standards and corresponding indications with associated measurement uncertainties and, in the second step, uses this information to establish a relation for obtaining a measurement result from an indication
Calibration curve	Expression of the relation between indication and corresponding measured value
Calibrator Calibration Standard	Realization of the definition of a given quantity, with stated value and associated measurement uncertainty, used as a reference
Certified Reference Material (CRM)	Reference material, accompanied by documentation issued by an authoritative body and providing one or more specified property values with associated uncertainty and traceability, using valid procedures
Characteristic	An entity or concept that is a distinctive or specific feature of a method, such as robustness or accuracy
Combined standard uncertainty	Standard measurement uncertainty that is obtained using the individual standard measurement uncertainties associated with the input quantities in a measurement model
Conformance probability	Probability that an item fulfills a specified requirement
Conformity assessment	Activity to determine whether specified requirements relating to a product, process, system, person, or body are fulfilled

Term	Definition
Consumer's risk	Probability that a particular accepted item is non-conforming
Conventional value	Value attributed by agreement to a quantity for a given purpose
Correction	Compensation for an estimated systematic effect
Coverage factor	Number larger than one by which a combined standard measurement uncertainty is multiplied to obtain an expanded measurement uncertainty
Coverage interval	Interval containing the set of true values of a measurand with a stated probability, based on the information available
Coverage probability	Probability that the set of true values of a measurand is contained within a specified coverage interval
Decision rule	Documented rule that describes how measurement uncertainty will be accounted for about accepting or rejecting an item, given a specified requirement and the result of a measurement
Definitional uncertainty	Component of measurement uncertainty resulting from the finite amount of detail in the definition of a measurand
Effect	ANOVA: contribution of a specific factor level to the output variable
Error	Measured value minus a reference value
Estimator	In statistics, it is a rule that creates an estimate from observed data. For example, the sample mean is an estimator for the theoretical mean
Evaluation/Characterization	Study of the qualities of a process, technique, or instrument to specify its characteristics and its adaptation to the intended purpose
Expanded uncertainty	Product of a combined standard measurement uncertainty and a factor larger than the number one
Fitness-for-purpose	Concept covering the level of appreciation of the ability of the data obtained by a measurement process to enable a user to make technically and administratively sound decisions
Guard band	Interval between a tolerance limit and a corresponding acceptance limit
Input quantity	Quantity that must be measured, or a quantity, the value of which can be otherwise obtained, to calculate a measured value of a measurand
Inspection	Conformity assessment by observation and judgment accompanied, as appropriate, by measurement, testing, or gauging
Interlaboratory study	Organization, execution, and evaluation of measurements or tests on the same or similar entities by two or more laboratories under predetermined conditions

Term	Definition
Intermediate precision condition	Condition of measurement, out of a set of conditions that includes the same measurement procedure, same location, and replicate measurements on the same or similar objects over an extended period, but may include other conditions involving changes
Intermediate precision variance	Measurement precision under a set of intermediate precision conditions of measurement
International System of Units	System of units, based on the international system of quantities, their names, and symbols, including a series of prefixes and their names and symbols, together with rules for their use, adopted by the General Conference on Weights and Measures (CGPM)
Interval	Interval is used together with the symbol $[a, b]$ to denote the set of real numbers x for which $a \leq x \leq b$, where a and b are real numbers. The term is used here for closed intervals. The symbols a and b denote the endpoints of the interval $[a, b]$
Limit of detection	Measured value, obtained by a given measurement procedure, for which the probability of falsely claiming the absence of a component in a material is, given a probability of falsely claiming its presence
Limit of quantification (1)	Smaller and/or greater concentration of the analyte that can be quantified, under the experimental conditions described in the method
Limit of quantification (2)	Smaller and/or larger quantity of the analyte in a sample that can be dosed under the experimental conditions described with defined uncertainty. It corresponds to the smallest and/or largest concentration of the validated range
Limit of quantification (3)	The lowest concentration of an analyte that can be quantified in a sample with acceptable risk of error, under indicated operating conditions
Limit of quantification (4)	The lowest concentration for which a coefficient of variation of repeatability equal to a given threshold is obtained
Linearity	Establishment that there is a linear relationship between the quantities found (or quantified) in samples and their reference values
Matrix	A set of constituents forming the sample, other than the researched analyte. Matrix effects (or matrix interferences) reflect the possible influence the matrix constituents may have on the measuring device response
Maximum permissible error	For a measuring instrument, maximum difference, permitted by specifications or regulations, between the instrument indication and the quantity being measured

Term	Definition
Measurand	Quantity intended to be measured (see Analyte)
Measured value	Value representing a measurement result
Measurement	Process of experimentally obtaining one or more values that can reasonably be attributed to a quantity
Measurement capability index	Tolerance divided by a multiple of the standard measurement uncertainty associated with the measured value of a property of an item. Generally taken to be four
Measurement procedure	Detailed description of a measurement according to one or more measurement principles and to a given measurement method, based on a measurement model, and including any calculation to obtain a measurement result
Measurement uncertainty (MU)	Non-negative parameter characterizing the dispersion of the values being attributed to a measurand based on the information used
Measurement method	Generic description of a logical organization of operations used in a measurement
Measurement model	Mathematical relation among all quantities known to be involved in a measurement
Measurement result	Set of values being attributed to a measurand together with any other available relevant information
Measuring interval or working interval	Set of values of quantities of the same kind that can be measured by a given measuring instrument or measuring system with specified instrumental uncertainty, under defined conditions
Measuring transducer	Device, used in measurement, that provides an output quantity having a specified relation to the input quantity
Nominal property	Property of a phenomenon, body, or substance, where the property has no magnitude
Output quantity	The measured value of which is calculated using the values of input quantities in a measurement model
Parameter	Statistical model coefficients or combination of coefficients are used to convey the performance of a method. They are random variables
Precision	Closeness of agreement between indications or measured values obtained by replicating measurements on the same or similar objects under specified conditions
Primary measurement procedure	Reference measurement procedure used to obtain a measurement result without relation to a measurement standard for a quantity of the same kind
Procedure	Operations to be carried out, precautions to be taken, and measures to be applied are contained in documents specific to each laboratory
Producer's risk	Probability that a conforming item will be rejected based on a future measurement result